

Smart School Management System (SSMS)

Project Documentation

A Windows Forms (.NET) Desktop Application

Visual Programming

1. Introduction

The Smart School Management System (SSMS) is a desktop application built using C# and Windows Forms (.NET) to help schools digitize and simplify their day-to-day administrative work. The system supports multiple types of users, including the Principal and Teachers, by giving each of them a dedicated dashboard and a set of tools relevant to their role.

Traditionally, schools rely on paper-based registers and manual processes for tasks such as registering teachers and students, publishing notices, tracking attendance, recording exam marks, and managing timetables. These manual methods are time-consuming, prone to human error, and make it difficult to retrieve information quickly. SSMS addresses these problems by providing a centralized, easy-to-use desktop interface where authorized users can log in securely and perform their tasks electronically.

The application follows a simple and consistent navigation flow: a loading screen is shown while the system initializes, after which the user is directed to a Sign-In page. Once authenticated, the user is taken to a role-specific dashboard, either the Principal's Dashboard or the Teacher's Dashboard, from which they can access the relevant modules such as Notice Create, View Notice, Exam Results, Student Attendance, Add Marks, Teacher Register, Student Register, and Timetable.

This document provides an overview of the project and describes its main functionalities with representative interface screenshots (shown here as placeholders).

1.1 Technology Stack

- Programming Language: C#
- UI Framework: Windows Forms (WinForms)
- IDE: Visual Studio
- Architecture: Multi-form desktop application with role-based navigation (Loading → Sign-In → Dashboard → Modules)

2. Main Functionalities

SSMS is composed of several forms (modules), each handling a specific part of the school management workflow. The main functionalities are described below, along with placeholder screenshots representing each interface.

2.1 Loading Page

When the application starts, a Loading page is displayed first. It simulates the system initializing by showing a progress bar and status messages (e.g., "Connecting to database...", "Fetching user data...", "Initializing components...") with a live percentage indicator. Once loading reaches 100%, the form automatically closes and redirects the user to the Sign-In page.

- Animated progress bar with percentage display
- Sequential status messages during initialization
- Automatic transition to the Sign-In page on completion

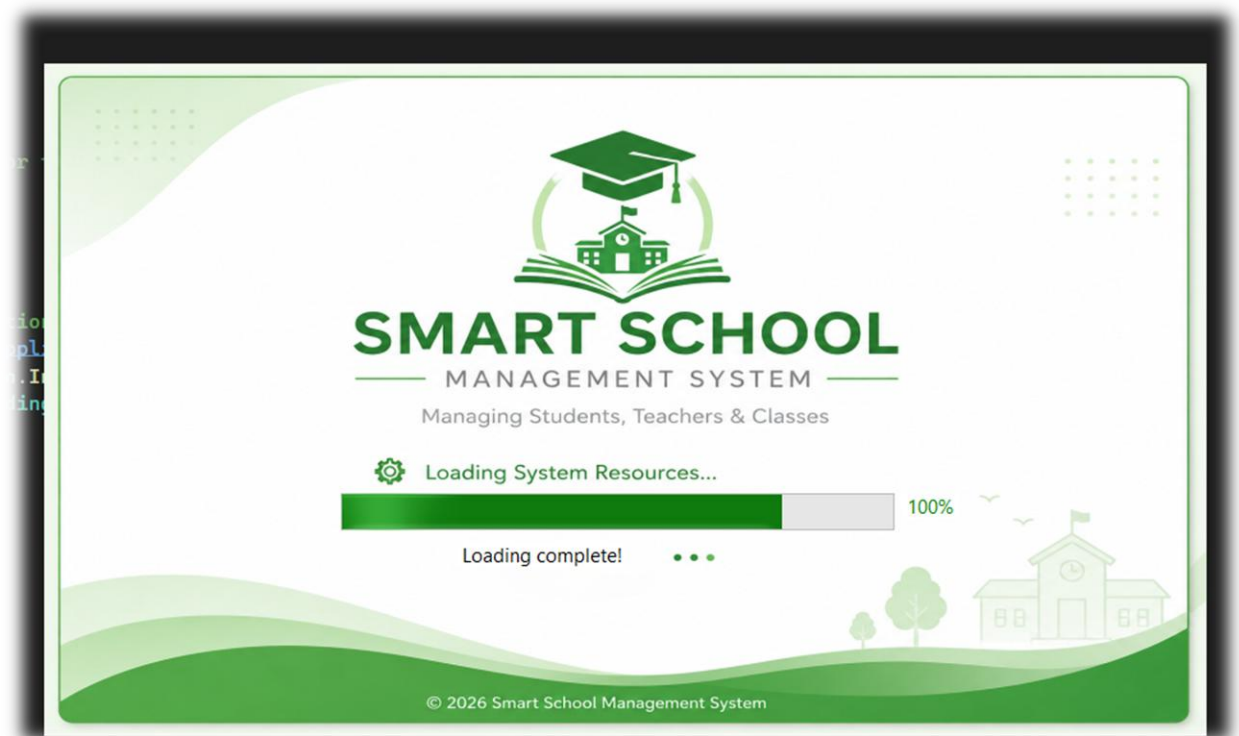


Figure 2.1 – Loading Page

2.2 Sign-In Page

The Sign-In page is the entry point for all users. It allows a registered user (Principal or Teacher) to enter their username and password to access the system, with a "Remember Me" option for convenience.

- Username and password input fields
- Remember Me checkbox
- Sign-in button that validates credentials and routes the user to their dashboard

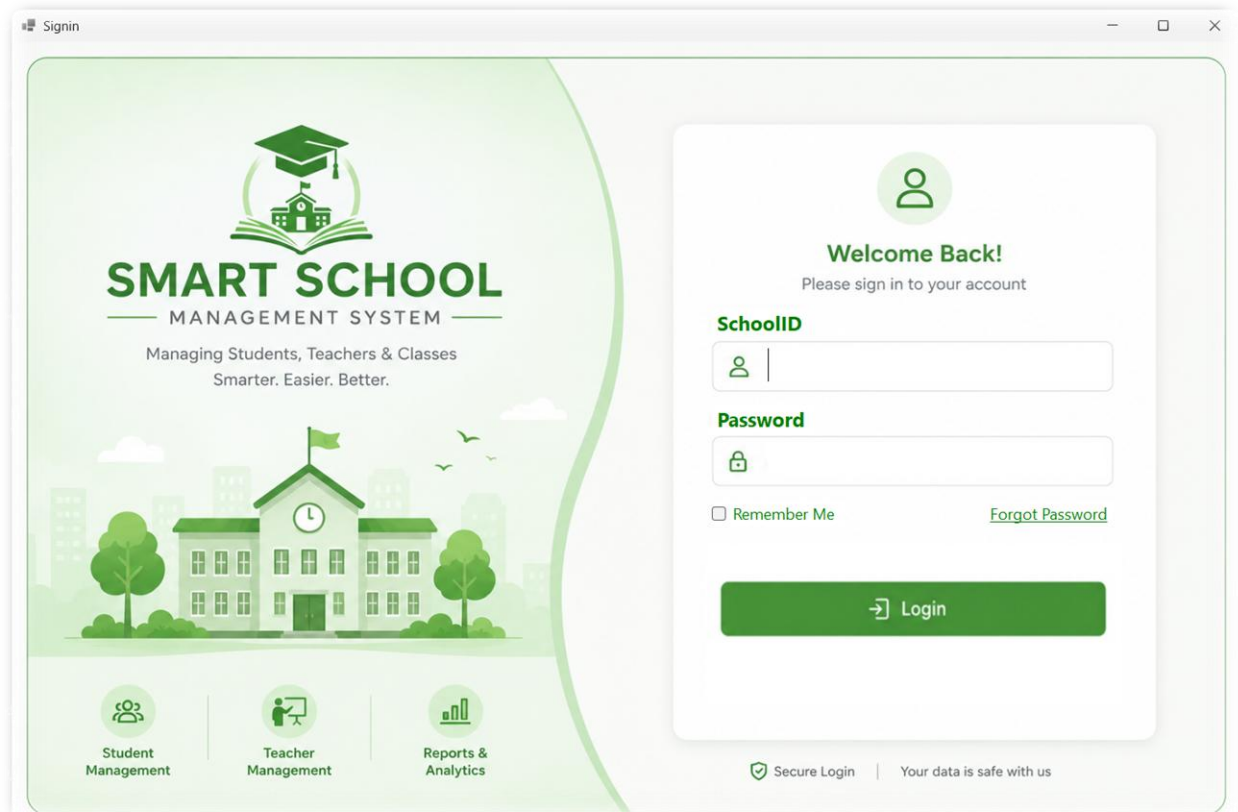


Figure 2.2 – Sign-In Page

2.3 Principal's Dashboard

After logging in as a Principal, the user is directed to the Principal's Dashboard. This dashboard acts as the central hub for administrative tasks, providing quick-access navigation to modules such as Teacher Register, Student Register, Timetable, and Notice Create.

- Sidebar navigation with quick-access buttons to all administrative modules
- Profile / welcome panel for the logged-in Principal
- Central hub linking to Teacher Register, Student Register, Timetable, and Notice Create

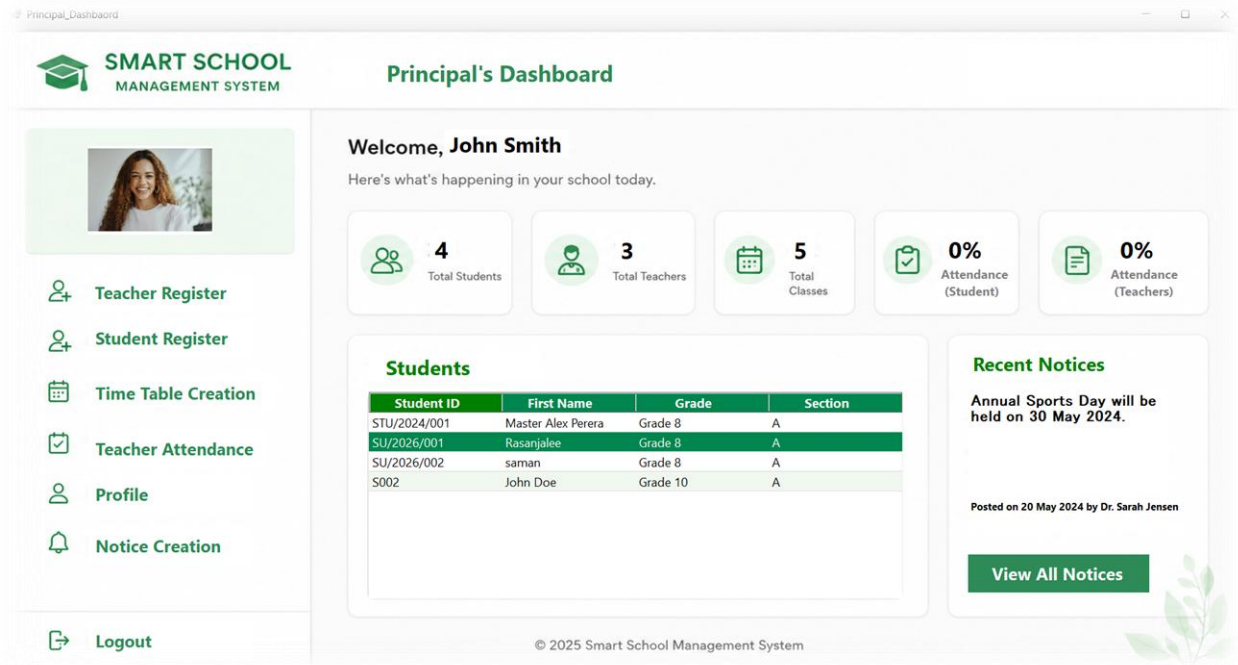
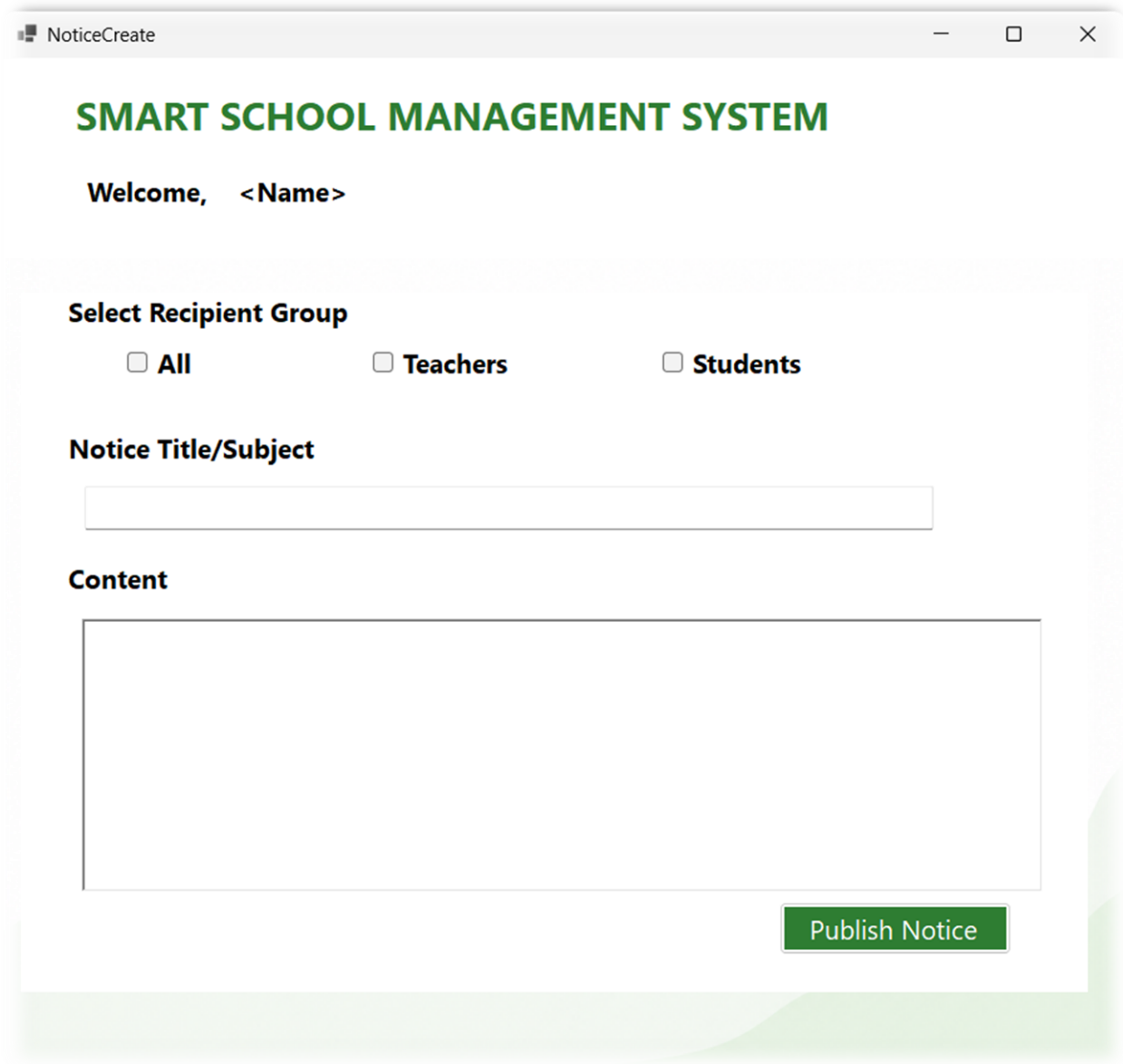


Figure 2.3 – Principal's Dashboard

2.4 Notice Create

The Notice Create module allows authorized staff to publish notices/announcements for the school. The user enters a notice title, types the notice content in a rich text box, selects the notice date using a date picker, and then publishes it.

- Notice title text field
- Rich text box for the notice body/content
- Date picker for the notice date
- "Publish" and "Clear" action buttons



The screenshot shows a web browser window titled "NoticeCreate". The page has a green header with the text "SMART SCHOOL MANAGEMENT SYSTEM". Below the header, there is a welcome message "Welcome, <Name>". The main content area is a light gray box containing a form. The form has a section titled "Select Recipient Group" with three radio buttons: "All", "Teachers", and "Students". Below this is a section titled "Notice Title/Subject" with a text input field. Underneath is a section titled "Content" with a large text area. At the bottom right of the form is a green button labeled "Publish Notice".

NoticeCreate

SMART SCHOOL MANAGEMENT SYSTEM

Welcome, <Name>

Select Recipient Group

☐ All ☐ Teachers ☐ Students

Notice Title/Subject

Content

Publish Notice

Figure 2.4 – Notice Create

2.6 Exam Results

The Exam Results module allows users to view and manage student examination results, helping teachers and administrators keep track of student academic performance.

- Search/filter results by student, class, or subject
- Tabular display of exam scores
- Option to view detailed results per student

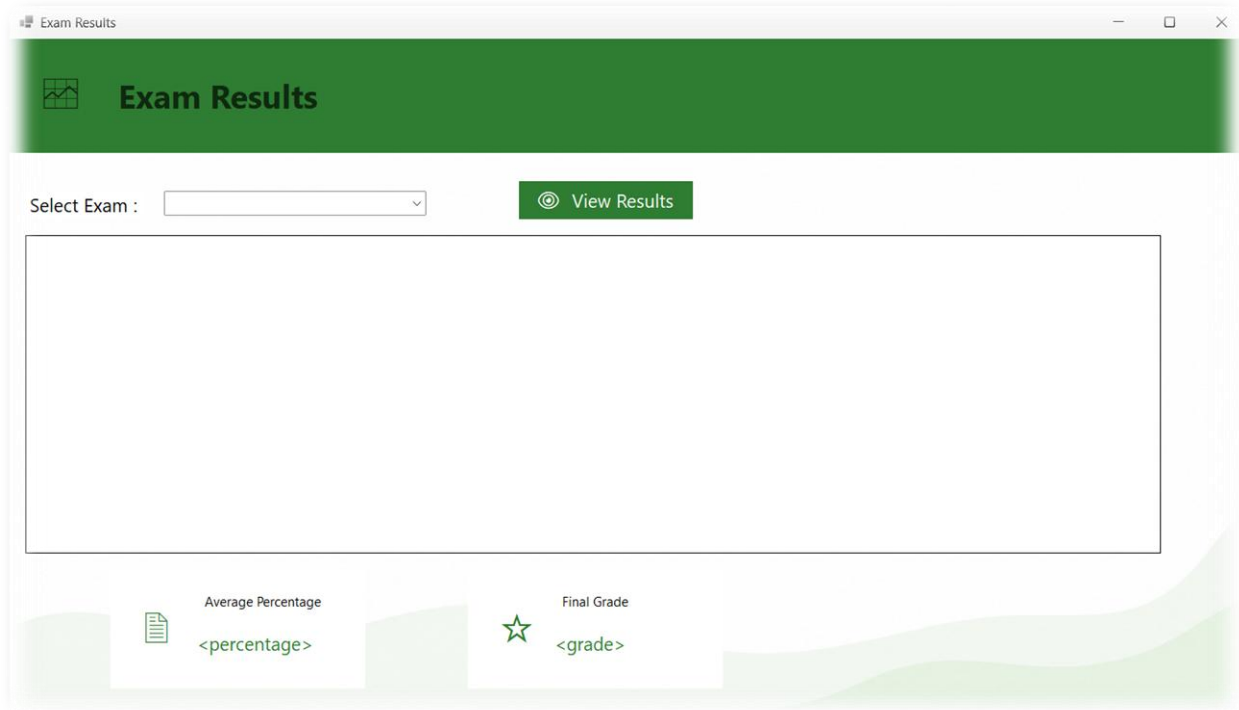


Figure 2.6 – Exam Results

2.7 Teacher's Dashboard

After logging in as a Teacher, the user is directed to the Teacher's Dashboard. This dashboard provides quick-access navigation buttons to teacher-related modules such as Student Attendance, Add Marks, and View Notice.

- Sidebar navigation with multiple quick-access buttons
- Profile / welcome panel showing the logged-in teacher
- Central hub linking to Student Attendance, Add Marks, and other teacher modules

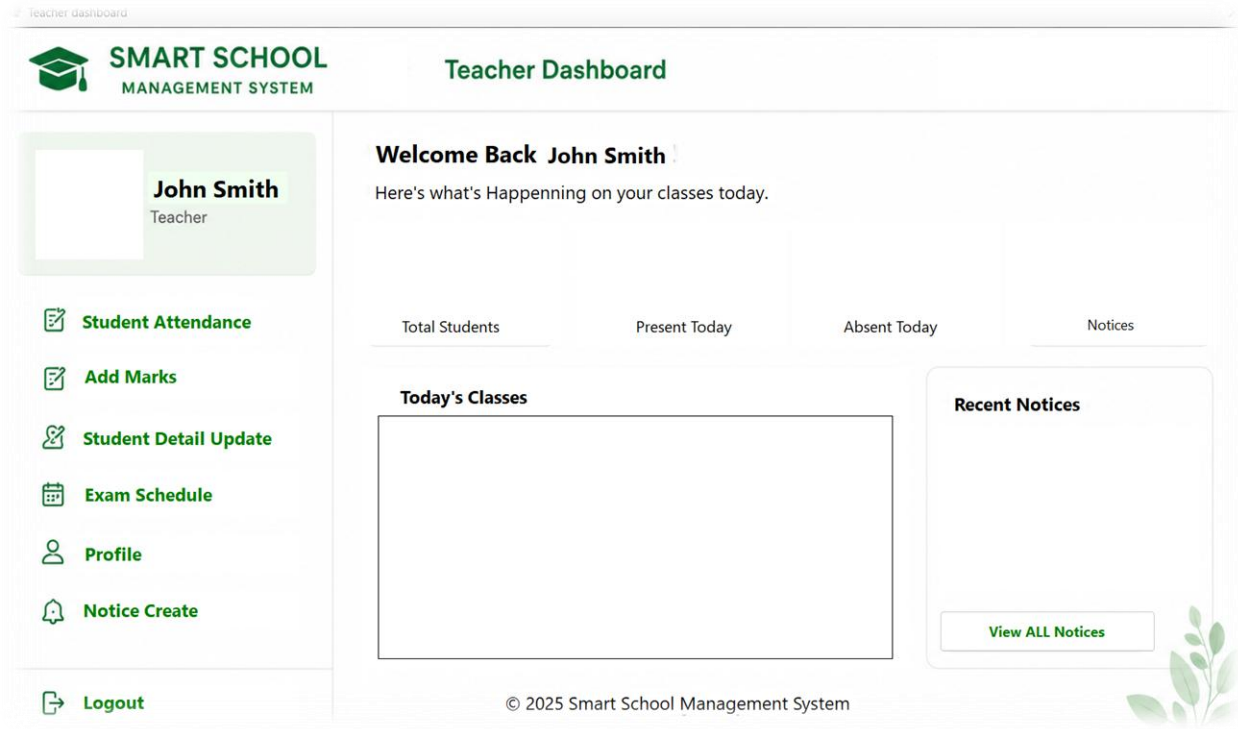
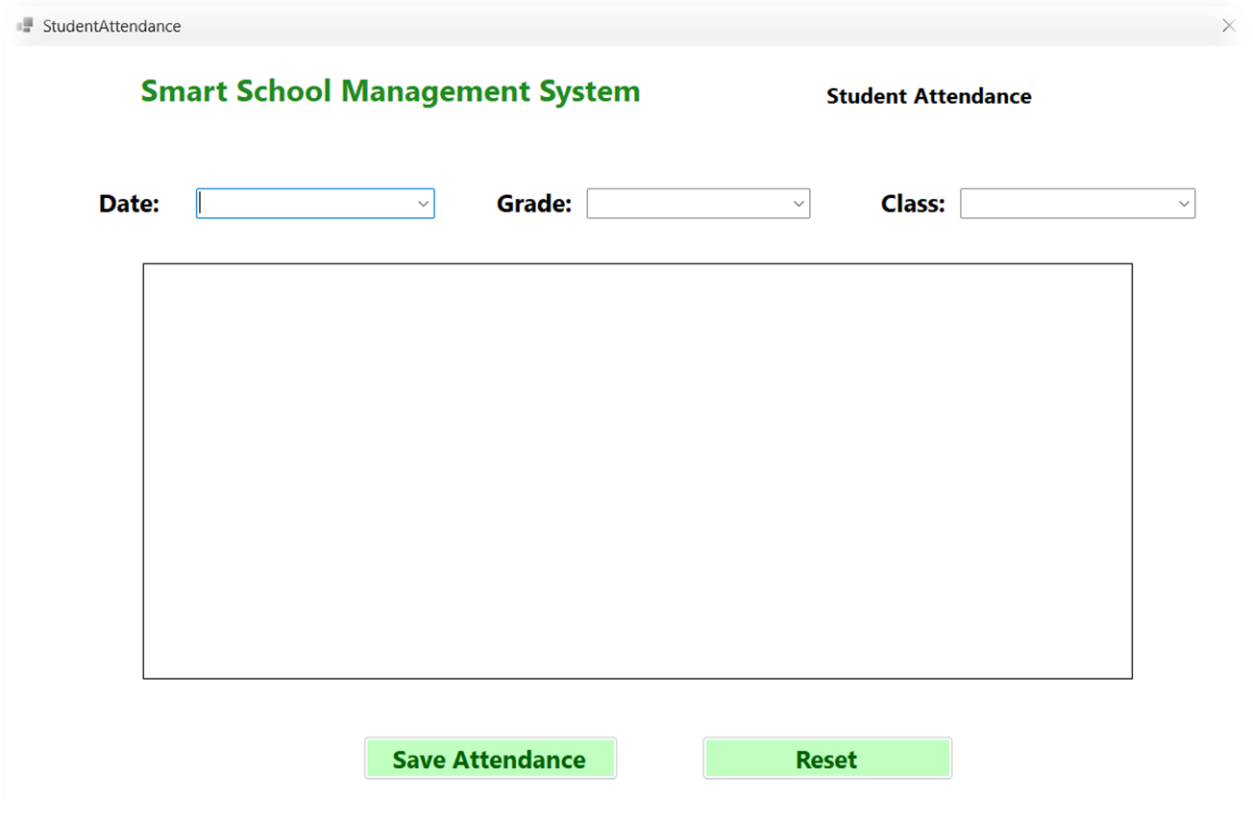


Figure 2.7 – Teacher's Dashboard

2.8 Student Attendance

The Student Attendance module allows teachers to mark and manage daily student attendance for their classes, replacing manual paper-based attendance registers.

- List of students per class for marking attendance
- Present/Absent selection options
- Save/submit attendance records



The screenshot shows a web application window titled "StudentAttendance" with a close button (X) in the top right corner. The main header area contains "Smart School Management System" in green text on the left and "Student Attendance" in black text on the right. Below the header, there are three dropdown menus labeled "Date:", "Grade:", and "Class:". The "Date:" dropdown is highlighted with a blue border. Below these fields is a large, empty rectangular box for a table or list. At the bottom of the form, there are two green buttons: "Save Attendance" and "Reset".

Figure 2.8 – Student Attendance

2.9 Add Marks

The Add Marks module allows teachers to enter and update examination or assignment marks for students, which then feed into the Exam Results module.

- Student and subject selection fields
- Marks input field
- Save/update functionality for entered marks

 addmarks ×

Add Student Marks

Grade:

Subject:

Exam:

Save Marks

Figure 2.9 – Add Marks

2.10 Teacher Register

The Teacher Register module allows the Principal to register new teachers into the system, including generating a unique Teacher ID and selecting relevant details such as subject/department.

- Input fields for teacher personal and professional details
- "Generate ID" button to auto-create a unique Teacher ID
- Dropdown selection for category/subject
- "Register Teacher" button to save the new teacher record

The screenshot shows a web browser window titled "Teacher Registration". The main heading is "Teachers Registration" in green. Below the heading are two tabs: "Personal Info" (active) and "Contact Info". The form is divided into three columns. The first column contains "First Name" (text input), "Date of Birth" (date picker showing "Sunday , June 21, 2026"), and "Gender" (radio buttons for "Male" and "Female", with "Male" selected). The second column contains "Last Nme" (text input), "NIC" (text input), and "Address" (text input). The third column contains "Profile Image" with a large empty box for the image. At the bottom center is a green "Next" button.

Figure 2.10 – Teacher Register

2.11 Student Register

The Student Register module allows the Principal to register new students into the system, capturing their personal details and assigning them to a class.

- Input fields for student personal details
- Class/grade selection
- "Register Student" button to save the new student record

Student Registration

Personal Info | Guardian Details | Academic | Account

Student First name

Student Last name

NIC

Date of Birth

Gender

Address

Profile Image

Next

Figure 2.11 – Student Register

2.12 Timetable

The Timetable module allows the Principal to create and manage class timetables, helping organize subjects and periods across the school week.

- Timetable grid showing days and periods
- Subject/teacher assignment per period
- Save/update timetable functionality

Timetable Creation - Relief Periods

	Time	Subject	Teacher	Room

Grade
Subject
Teacher
Start Time
End Time

Day
Room

Save

Figure 2.12 – Timetable

Contribution of Members

1. PS/2022/005 – W.M.T.N. Fernando
 - Loading page
 - Sign-in page
 - Principle's dashboard
2. PS/2022/011 – P.G.K.M. Bandara
 - Notice create
 - View notice
 - Exam results
 - Upload assessment
3. PS/2022/024 – U.S.C. Bandara
 - Teacher's dashboard
 - Student attendance
 - Add marks
 - Student details update
 - Exam schedule

4. PS/2022/116 – A.A.B. Gimhani

- Teacher register
- Student register
- Timetable

Details of Group Members

#	Student Number	Full Name (Initials)	Role
1	PS/2022/005	W.M.T.N Fernando	Group Leader
2	PS/2022/011	P.G.K.M Bandara	Member
3	PS/2022/024	U.S.C Bandara	Member
4	PS/2022/116	A.A.B Gimhani	Member

GITHUB LINK - <https://github.com/Thisaru2001/SSMS.git>